DSA REFERENCE

Sliding Window Problem Ladder

A sequenced practice ladder from warm-up to interview stretch.

Represent a contiguous range with incrementally maintained state instead of recomputing every candidate.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

DSA / SLIDING-WINDOW

DSA

Sliding Window Problem Ladder

CORE IDEA

Represent a contiguous range with incrementally maintained state instead of recomputing every candidate.

Recognition cues

+ contiguous substring or subarray
+ longest or shortest valid range
+ fixed-size range
+ state can be updated at both ends

Implementation sequence

+ identify fixed versus variable length
+ define validity
+ add right once
+ shrink with a while loop
+ record at the correct moment

Invariant

The state describes exactly the inclusive range from left through right, and the shrink loop restores validity.

Worked reasoning

For longest substring with at most k distinct values, add right, shrink while distinct count exceeds k, then record length.

Complexity

O(n) because each element enters and leaves the window at most once; state is usually O(k) or O(alphabet).

Failure modes

- [ ] using if instead of while
- [ ] stale counts
- [ ] recording before restoring validity
- [ ] using monotonic-sum logic with negative values

Practice ladder

Sliding Window Problem Ladder: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Maximum Average Subarray I**	State the cue, invariant, edge case, and proof before implementation.
2	**Longest Substring Without Repeating Characters**	State the cue, invariant, edge case, and proof before implementation.
3	**Minimum Size Subarray Sum**	State the cue, invariant, edge case, and proof before implementation.
4	**Permutation in String**	State the cue, invariant, edge case, and proof before implementation.
5	**Minimum Window Substring**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS

REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.